Sets

```
collections of mutable, unordered, un-indexed distinct elements, store only
unique items
# Cannot store mutable elements, like lists, dictionaries, sets, because
sets are implemented by hash table, find items really fast
```

Creating a Set

```
set()
    OR
{" ", " ", " "}
```

Length of a Set

```
len() method
```

Check Item in a Set

```
item in set # True/False
    OR
set1.isdisjoint(set2) # Returns True if both sets does not share any element
```

Adding Items to a Set

```
- set.add(item) -> add one item at a time
- set.update([list]) -> add multiple items
```

Removing Items from a Set

```
- set.remove(item) -> if not found, raise errors
- set.discard(item) -> if not found, not raise error
- set.pop() -> remove random item, use print() to return removed item
```

Clearing the Set

```
set.clear()
```

Deleting the Set

```
del set
```

Convert List to Set

```
set = set(list) -> remove duplicated, order willl be random
```

Joining Sets

```
- set3 = set1.union(set2) -> return a new set
- set1.update(set2) -> insert set to a given set
```

Finding Intersection of Items

```
set1.intersection(set2) -> return set of items which are in both sets
```

Checking Subset and Superset

```
- issubset() method
    set1.issubset(set2) -> True/False
- issuperset() method
    set1.issuperset(set2)
```

Difference Between Two Sets

```
- set1.difference(set2) -> present in set 1 but not in set 2
- set1.symmetric_difference(set2) ->  present in either of the sets, but not
in both sets
```